

VU TRAN ANH MINH

Final year student at RMIT University

PROFILE

Final-year student in Robotics and Mechatronics Engineering at RMIT University, eager to apply for the Production intern position. Passionate about collaborative teamwork, delivering high-quality outcomes, and embracing new challenges. While I may not meet every technical requirement yet, I bring a strong work ethic, a willingness to learn, and an adaptable mindset. I'm excited for the opportunity to gain hands-on engineering experience and grow under the guidance of industry professionals.

EDUCATION

RMIT VIETNAM, Ho Chi Minh City.
Bachelor of Engineering (Robotics and Mechatronics Engineering)— [2022–2026]
Relevant coursework:

- ·Engineering Mathematics; Engineering Science; Engineering Mechanics; Design for Assembly and Automation;

Advanced Manufacturing and Mechatronics: What We Make and How We Make It

PROJECTS

Bridge Design & Stress Analysis

- Collaborated with a team to design and construct a small-scale functional bridge.
- Performed stress analysis and structural calculations using core mechanical engineering principles.
- Focus Theories: Static equilibrium equations, internal force distribution, stress-strain relationships, material strength under axial and bending loads.

Chariot Transport Robot

- Assisted in developing a lightweight-object transport chariot robot.
- Contributed to mechanical sketching, prototyping, and testing.
- Focus Theories: Dimensional tolerance analysis, 3D printing design constraints, SolidWorks parametric modeling principles.

Autonomous Robot with Emotion & Color Recognition

- Built and programmed autonomous robots using ESP32 and ESP32-CAM.
- Integrated facial emotion recognition, voice command control, and visual path planning with computer vision.
- Tuned systems to detect colors, identify start/goal points, and map the environment for navigation.
- Focus Theories: Color filtering matrices, occupancy grid mapping, shortest path algorithms (e.g., Dijkstra or A*), pose estimation through time-based dead reckoning.

Mechanical Design

- Researched and applied legged locomotion principles to build a functional Theo Jansen mechanism.
- ·Involved in structural assembly.
- Focus Theories: Jansen linkage kinematics, mechanical link optimization, torque-to-motion transfer in planar linkages.

Vertical-Axis Hydrogenerator Design

- Developed detailed mechanical designs for the main drive shaft, bearing selection, keyed joint, and turbine-shaft coupling under specified load and speed conditions.
- Conducted engineering analysis on torque transmission, material selection, and structural integrity while optimizing for weight and manufacturability.
- Produced CAD models and technical documentation in accordance with professional engineering standards.
- Focus Theories: Torque transmission theory, shear stress in keyed joints, fatigue safety factors, shaft design under bending and torsion.

Robot Arm Kinematics Simulation

- Performed joint and link coordination simulations to control a robot arm using transformation matrices.
- Visualized arm movement, velocity, and position through kinematic equations and forward transformation principles.
- Focus Theories: Homogeneous transformation matrices, Denavit-Hartenberg parameters, joint-angle to end-effector mapping.

WORK EXPERIENCES

Private Tutor – Ông Mặt Trời Tutoring Center

- Provided one-on-one tutoring sessions in [subjects you taught, e.g., Math, Physics, English] for students from grade 5 to 9
- Customized lesson plans based on student ability and progress, helping improve academic performance.
- Built strong rapport with students and parents, fostering a supportive learning environment.

EXTRACURRICULAR ACTIVITIES

Outgoing Volunteer Service Member – AIESEC Vietnam - 2023

- Advised and scheduled volunteer exchange projects for customers in collaboration with international partner entities.
- Coordinated activities and communication across cultures to ensure smooth delivery of leadership-oriented volunteer programs.

Logistics Assistant – RMIT Japanese Cultural Club - 2022

- Assisted in managing storage accessibility and event logistics.
- Supported the transportation and handling of goods and materials for internal and public club events.

Project Leader – Robotics Project, RMIT Neo Culture Tech Club - 2025

- Led a team to develop a functional robotic system as part of a cultural-tech initiative.
- Defined project scope, delegated tasks, and tracked team progress to ensure timely and effective project delivery.

AWARDS

TOP 16 - NEO LEAGUE

SKILLS

- Engineering & Technical: SolidWorks (Intermediate), Stress Analysis, Embedded Systems (ESP32, ESP32-CAM), Control Systems, Theo Jansen Mechanism, Robot Arm Kinematics
- Programming: C++ (Intermediate), Java (Intermediate), Python (Basic), Arduino IDE
- Automation & Computer Vision: Mapping Algorithms, Pathfinding (Dijkstra/A*), Color Filtering, Motion Planning, PLC/SCADA (Basic understanding)
- Digital Tools: Excel (Intermediate), Power BI (Basic), Power Apps (Basic), Git
- Soft Skills: Team Collaboration, Adaptability, Fast Learner, Strong Work Ethic, Problem-Solving, Leadership (Project Coordination)
- Languages: Vietnamese (Native), English (IELTS 7.0), Japanese (Entry Level)

CONTACTS

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